
B2Ai
VOiCE

2026 Voice AI Hackathon

Logistics, Overview, and Data Access

Andrea Krussel

WashU Medicine

Training, Recruitment, and Mentoring (TRM)

B2AI VOICE

Agenda

1. About Bridge2AI
2. Event logistics
3. Challenge track scoring and instructions
4. About the data set
5. Challenges
6. Tools
7. Protocol
8. Teaming
9. Education

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Our voices.
Our health.
Our future.

Voice AI Hackathon Pre-Assessment

Enter the Survey Access Code

To allow a respondent to begin this survey, have them navigate to the URL below and enter the survey access code. The code is permanent and will never change. (Note: The web address is the same for all projects and surveys, so you may bookmark the address on a computer or device to quickly return to it multiple times.)

1.) Go to this web address:

https://redcap.wustl.edu/redcap/surveys/

2.) Then enter this code:

EJHXLX9MF

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The Bridge2AI Program

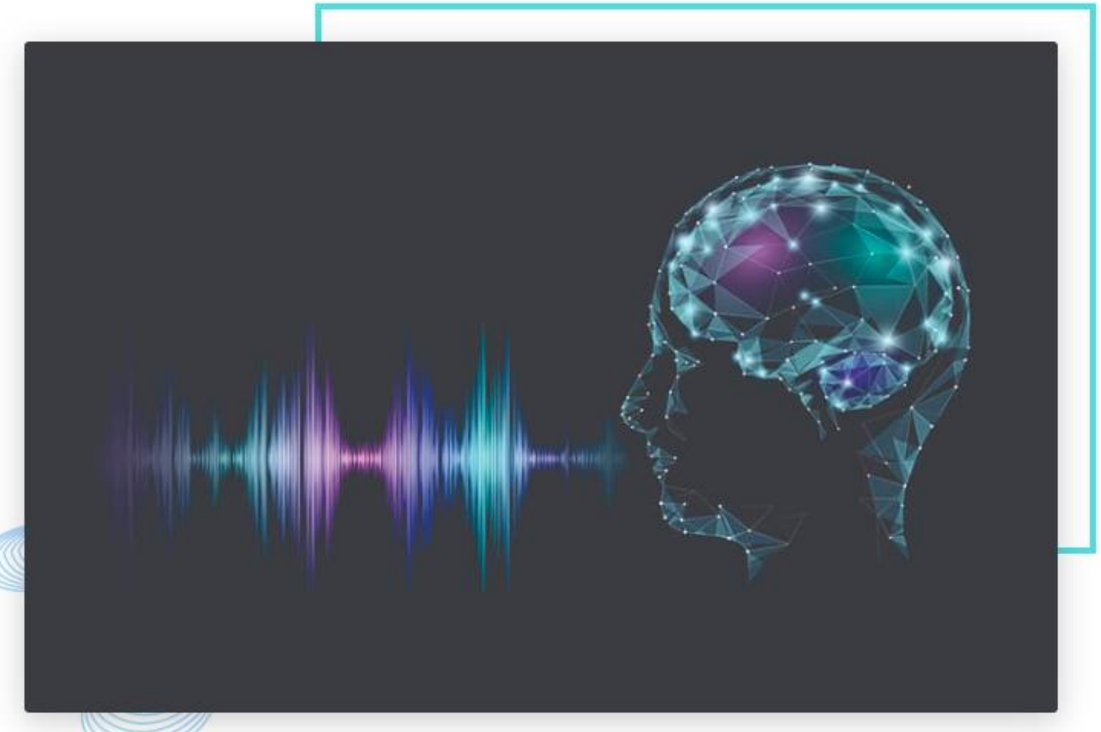
The NIH Common Fund's Bridge to Artificial Intelligence (Bridge2AI) program will propel biomedical research forward by setting the stage for widespread adoption of artificial intelligence (AI) that tackles complex biomedical challenges beyond human intuition. A key step in this process is generating new "flagship" data sets and best practices for machine learning analysis. Machine learning is a type of Artificial Intelligence (AI) that provides machines (like computers) the ability to automatically learn and improve from experience, without being explicitly programmed. The biomedical research community generates a wealth of data, but most of these data are not suitable for machine learning because they are incomplete.

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The Bridge2AI Program

By bringing technological and biomedical experts together with social scientists and humanists, the Bridge2AI program will help bring solutions to this deficit by:

- Generating new flagship biomedical and behavioral data sets that are ethically sourced, trustworthy, well-defined, and accessible
- Developing software and standards to unify data attributes across multiple data sources and across data types
- Creating automated tools to accelerate the creation of FAIR (Findable, Accessible, Interoperable, and Reusable) and ethically sourced data sets
- Providing resources to disseminate data, ethical principles, tools, and best practices
- Creating training materials and activities for workforce development that bridges the AI, biomedical, and behavioral research communities



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Bridge2AI Voice

Voice as a biomarker of health is one of 4 data generation projects funded by the **NIH's Bridge2AI program**.

Our group, **The Bridge2AI-Voice Consortium** aims to integrate the use of voice as biomarker of health in clinical care by generating a substantial multi-institutional, ethically sourced, and diverse voice database linked to multimodal health biomarkers to fuel voice AI research and build predictive models to assist in screening, diagnosis, and treatment of a broad range of diseases. Data collection will be made possible by software through a smartphone application linked to electronic health records (EHR) and other health biomarkers such as radiomics, and genomics, and supported by federated learning technology to protect data privacy.

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About the Hackathon

About the Hackathon

The Bridge2AI-Voice Hackathon is an exciting one-day event where participants will tackle real-world clinical and technical challenges using the Bridge2AI-Voice dataset. Registration is open to students, startups, and researchers and includes participation in the event, access to in-demand educational content, and mentorship from leaders in the field. This is your chance to showcase your AI and data skills in a collaborative, fast-paced environment using a voice AI dataset validated across four cohorts of diseases (neurological diseases, respiratory diseases, vocal diseases, and mood/psychiatric diseases)

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Event Logistics – Challenge Track

*

Challenge Track

- 8:30 am-9:00 am: Introduction, welcome, explanation of challenges and data access
- 9:00 am-1:30 pm: Work on challenges
- 1:30-2:30 pm: Create presentations
- 2:30-4:30 pm: Presentations and judging
  - 5 minutes per presentation plus Q&A
- 4:30-5:00 pm: Closing and awards

**Schedule subject to change*

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Event Logistics – Education Track

*

Education Track

- 8:30 am-9:00 am: Introduction, welcome, explanation of challenges and data access
- 9:00 am-10:00 am: Overview of Voice Biomarkers, Protocol development, role of the clinician, Maria Powell
- 10:00 am-11:00 am: How to use B2AI dataset and build a basic pipeline - Steven Bedrick
- 11:30 am - Noon : Overview of the Zana Platform, Julia Hoxa
- Noon - 2:30 pm: Participants can join teams to observe and learn, or can access our curriculum for self-guided learning
- 2:30 - 4:30 pm: Presentations and judging
  - 5 minutes per presentation plus Q&A
- 4:30-5:00 pm: Closing and awards

**Schedule subject to change*

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Judging Criteria

1= needs work - 5=exceptional quality

Methods – 5 points

Project Development

Were creative features or methodologies used in the model development?

Problem is clearly defined - what is the problem that is solved and how does it improve patient outcomes

Evaluation

Example: How well does the model perform on unseen data?

Are there measures in place to ensure fairness and equity in risk prediction?

Mechanics – Accuracy and Performance 5 points

Does the model accurately predict the risk of medical conditions?

How easy is it to understand the model's predictions?

Results – 5 points

Effectiveness

Are there measures in place to ensure fairness and equity in risk prediction

Could a final version of this project produce clinically meaningful and useful inputs for decisic making?

Bias Mitigation

Does the model consider and mitigate potential biases in the data?

Does the model uphold privacy and confidentiality standards for medical data?

Presentation/

Communication - 5 points

Approach and Insights

How effectively does the team communicate their approach, results, and insights?

Engagement with Judges · Does the team effectively address questions and feedback from the judges?

Is the model transparent and explainable to clinicians and stakeholders?

In Under Eight Slides Address the Following:

1. The Problem:
   a) What is the clinical story/scenario you are trying to tell?
   b) How many people does this affect?
2. What is the literature?
   a) Background information
   b) Limitations
3. Methods:
   a) What approach/data was used to build your model?
   b) What are the limitations of this model?
4. Results
5. Insights to consider:
   a) How is this AI Model different?
   b) How does it address Bias?
   c) Is there potential for misuse of this model/technology?
   d) What would you do if you had one more day to work on this model?

Access the PowerPoint template here:
https://wustl.box.com/s/9y7m7ulufbitc2ayoy2kk9uspfskxpi1

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Box for Presentations

Please drop your final presentations in the box below:

https://tinyurl.com/3yaf9n8h

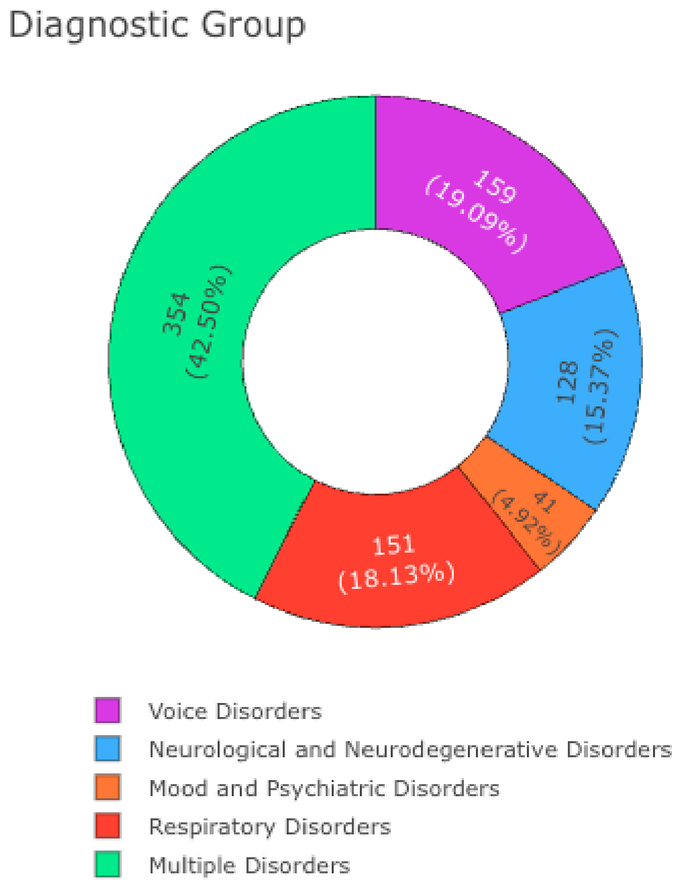
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Voice AI Dataset

Bridge2AI Voice Dataset

Flagship Dataset of Voice as a Biomarker of Health

B2AI-Voice is generating a robust, large-scale voice dataset to fuel the discovery of voice as a biomarker of health. Through collaborative research, the development of best practices, education, and events, we advance the science and application of voice AI to improve patient outcomes.



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Data Access

We ask that you obtain access to the Voice AI feature data set in advance of the hackathon by visiting this site:

https://physionet.org/content/b2ai-voice/3.0.0/

At the bottom of the page, you will be prompted to create or log in to a Physionet account, sign a data use agreement (DUA), and become a credentialed user, as illustrated below.

When registering as a credentialed user on Physionet and you are not a student or postdoc, please feel free to use Dr. David Dorr as a reference: David A. Dorr, MD, MS, Professor of Informatics, Clinical Epidemiology, and Translational Data Science, Department of Medicine, Oregon Health and Science University, dorrd@ohsu.edu or Dr. Steven D. Bedrick, Associate Professor, Informatics, Clinical Epidemiology, and Translational Data Science, Department of Medicine, Oregon Health and Science University, bedricks@ohsu.edu.

If you are unable to sign the DUA, please don't let that hinder you from attending. During the group breakout sessions, we can either help you register or pair you with a registered user.

B2AI VOICE

Documentation Site

https://b2ai-voice.org/bridge2ai-voice-dataset/

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Presentation of the Challenges

Challenge 1: Voice Across the Lifespan

As we age, our voices change: first at puberty, and then through adulthood. This has several implications for the clinical use of voice biomarkers:

Models trained on adult voices may not work as well when run against those of children

Longitudinal data within a single patient's voice may confound the signal of a particular disorder

During periods of rapid voice change (adolescence, gender transition, etc.), a patient's age by itself may not be a reliable yardstick for choosing between a pediatric and adult model.

Our dataset is unique in that it includes a significant number of both pediatric and adult subjects, making this a perfect opportunity to dig into this understudied area!

Challenge 1: Voice Across the Lifespan Ideas:

1

Quantify which features are most stable between pediatric and adult voices.

2

Look for tasks that are maximally stable across age groups for a given condition/s.

3

Focus on an age range of interest, e.g., early childhood through adolescence.

Challenge #2: Differential Diagnosis



Many disorders share acoustic signatures. Can you identify disorders that "sound" the most similar, and then figure out how to differentiate them?



Relatedly, some disorders that are *clinically* hard to distinguish from one another have acoustic signatures that could help in differentiation (e.g., benign vs. malignant vocal fold lesions).



Work with clinical partners to learn more about what might be clinically hard to diagnose, don't just guess from the data!

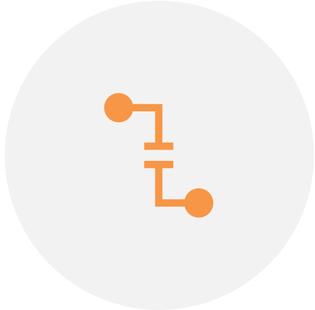
Challenge #3: Synthetic Data



Sharing sensitive information is subject to many different regulations around the world.



Can you build a corresponding synthetic dataset that is comparable to the original dataset from a statistical perspective?



This means that the individual, joint, and conditional distributions of features would be similar, without any row of data being identical to or proximal to the original dataset.

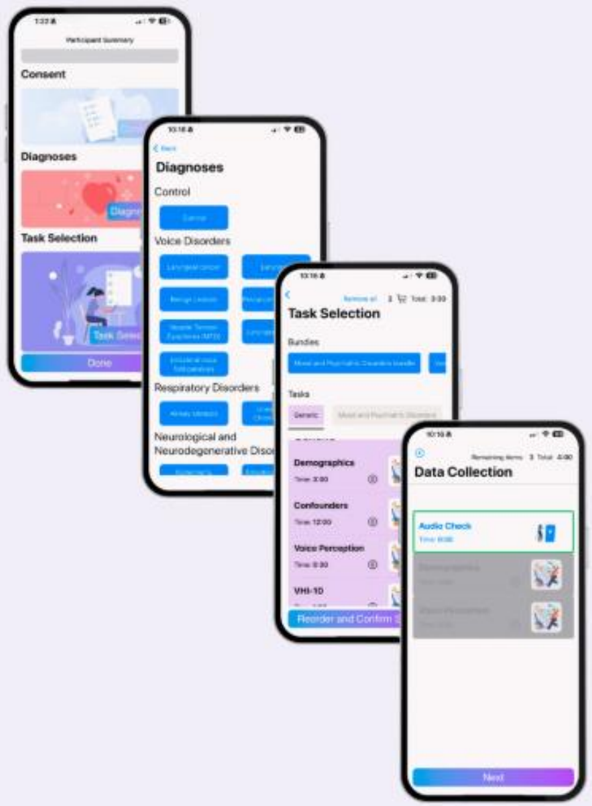
Data Access Tools

B2AI-Voice Tools & Resources

B2AI-voice has created a broad set of tools and resources to help researchers, educators, and clinicians make the most of the B2AI-Voice dataset. These include protocols, preparation kits, educational materials, ethics and governance resources, and scholar networks.

We also develop explainable AI (XAI) tools and best practices to ensure transparency, usability, and reproducibility in voice-based research. Together, these resources are designed to maximize the impact and accelerate the responsible use of the B2AI-Voice dataset across diverse domains.

https://b2ai-voice.org/tools-and-resources/



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Zana VOICE-BIOME Platform

We are pleased to share that our Hackathon sponsor Zana AI has provided access to their Voice AI platform for Healthcare for the challenges. Information on accessing Zana can be found via the links below:

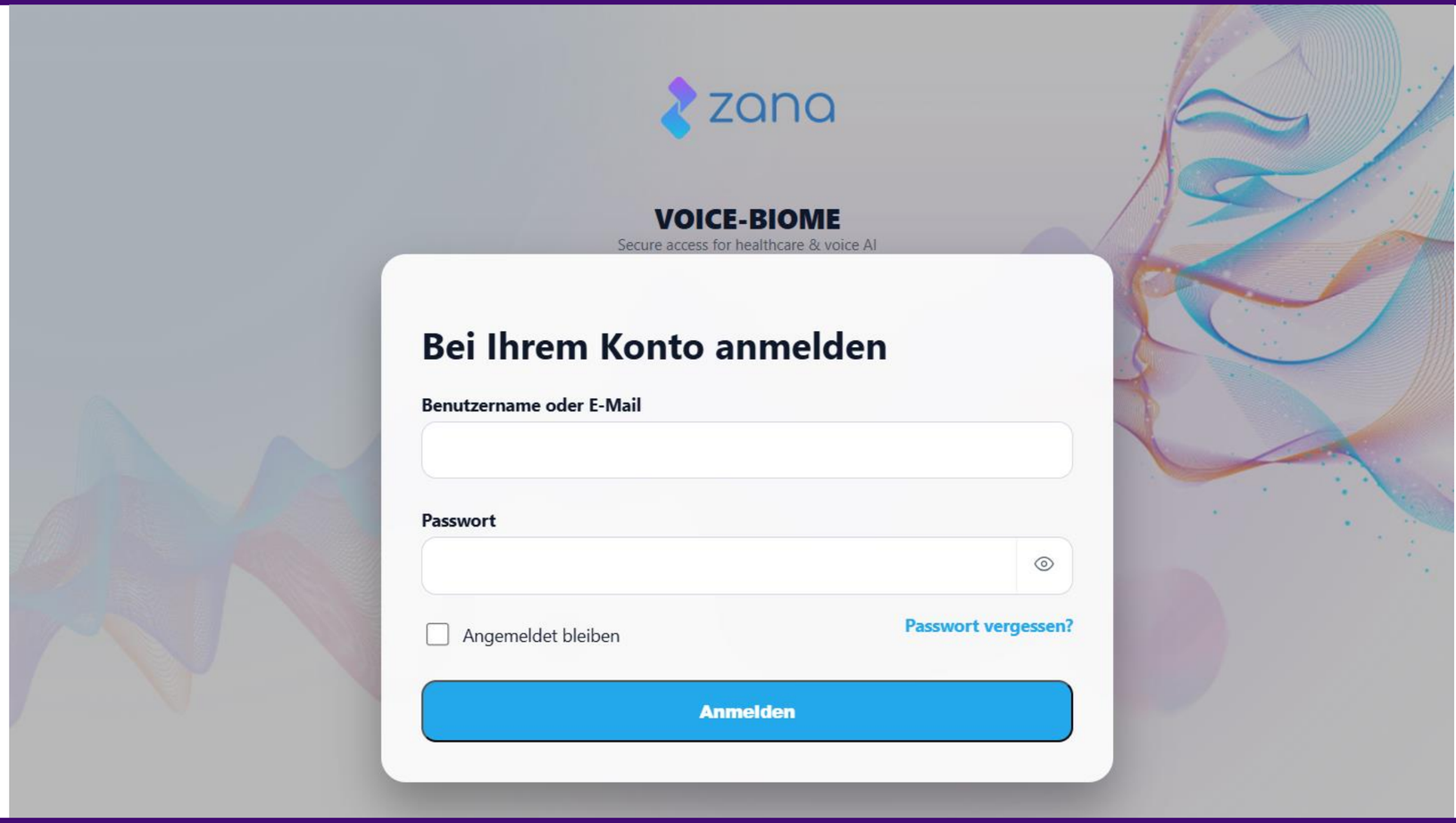
- Voice AI Help Desk: https://saas-dev.ex.zana.ai/auth/help-center
- Import the dataset (after signing DUA): https://saas-dev.ex.zana.ai/auth/help-center/hackathon/import-b2ai-dataset
- Analytics Dashboard: https://saas-dev.ex.zana.ai/auth/help-center/analytics-dashboard/reading-the-dashboard

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Zana VOICE-BIOME Platform

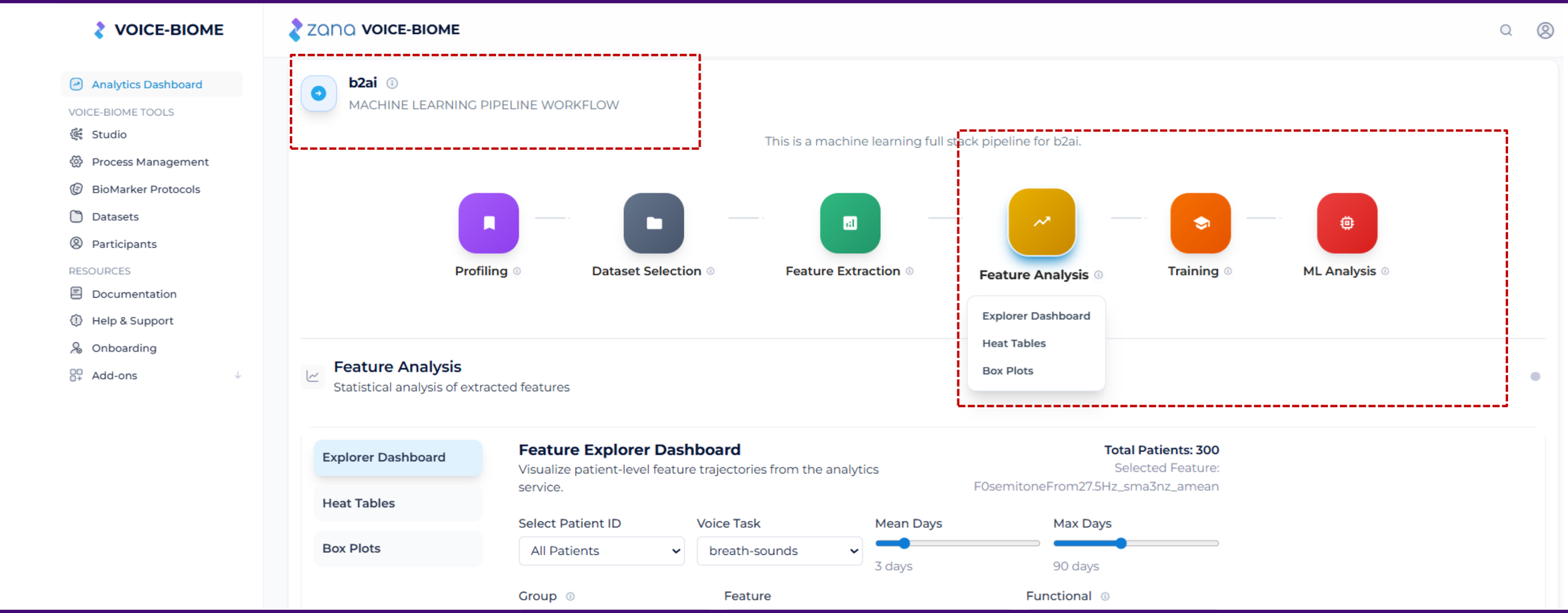
Request access to the platfrom.

You get a workspace dedicated to your team.



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Zana VOICE-BIOME Platform



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Teaming and Protocol

Maria Powell, PhD
Vanderbilt University Medical Center

JAMA Otolaryngology–Head & Neck Surgery

Viewpoint

Voice as an AI Biomarker of Health—Introducing Audiomics

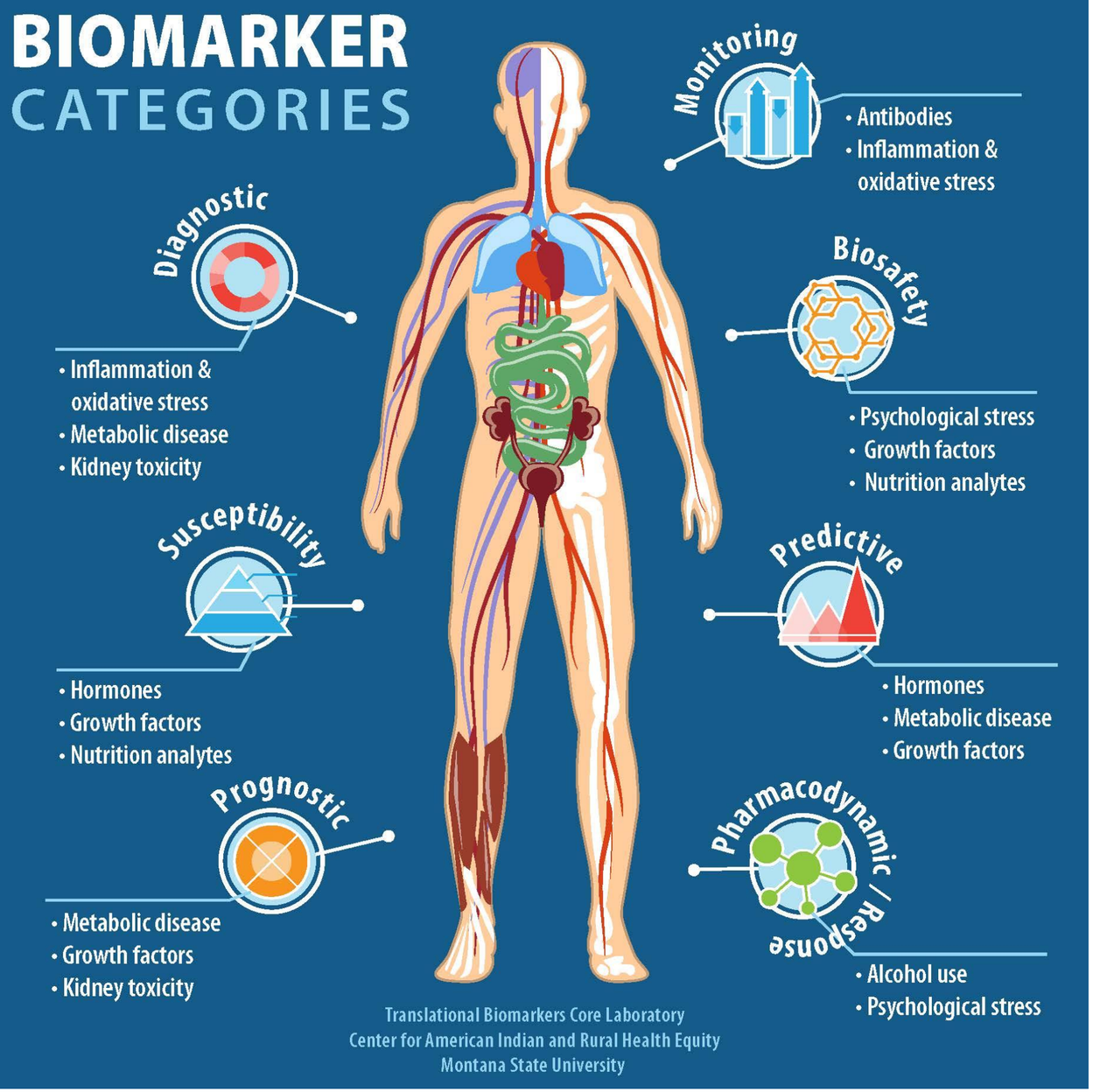
Yaël Bensoussan, MD, MSc[1]; Olivier Elemento, PhD[2]; Anaïs Rameau, MD, MPhil[3]

Author Affiliations | Article Information

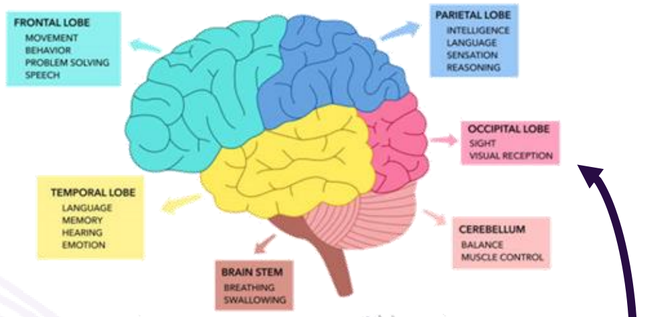
RELATED ARTICLES MEDIA

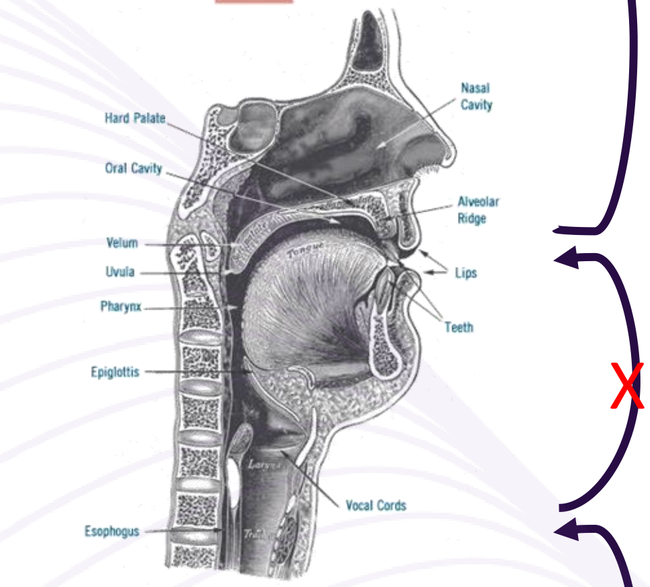
Interdisciplinary field of audio analysis applied to biomedicine to identify unique audio biomarkers of health and disease

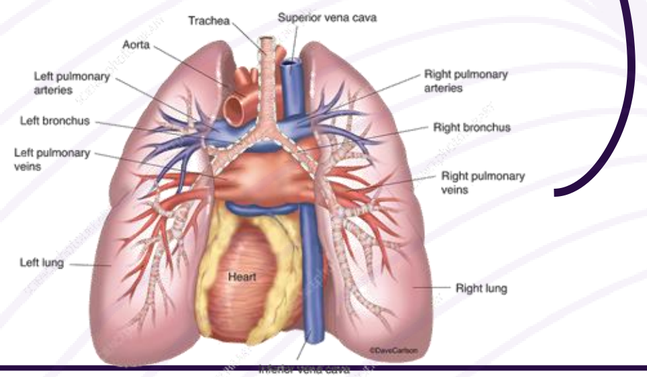
Biomarkers are indicators of a biological state that provide a standardized, valid, and precise way of evaluating environmental exposure, effect, or susceptibility, diagnosis, or progression of disease and infection. Biomarkers can be biological, physical, or chemical, and they are measurable parameters in an organism indicative of a specific biological state.



https://www.montana.edu/cairhe/translational-biomarkers-core/







X

- PEF
- PIF
- PEV1
- PVC
- X F(0)
- X Frequency range
- X CPP
- X Jitter
- X Shimmer
- X Voice onset time
- X Speech rate
- X MPT
- X MFDR

1 ORIGINAL PROTOCOL

PART A: All Participants	PART B: Mood Disorders	PART B: Respiratory Disorders	PART B: Voice Disorders	PART B: Neurological Disorders
About Me	**About Me**	**About Me**	**About Me**	**About Me**
Demographics	Mood-related Medical History	Respiratory Medical History	Voice Medical History	Neurological Medical History
General Medical History	Mood-related Confounders	Respiratory Confounders	Voice Confounders	Neurological Confounders
General Confounders				
Validated QOL Questionnaires	**Validated QOL Questionnaires**	**Validated QOL Questionnaires**	**Validated QOL Questionnaires**	**Other Tasks**
Voice Handicap Index 10	Positive and Negative Affect Scale	Dyspnea Index	Voice Problem Severity	Winograd task
Patient Health Questionnaire-9	Custom Affect Scale	Leicester Cough Questionnaire (LCQ)		
GAD-7 Anxiety	DSM-5 Adult		**Speech Tasks**	**Speech Tasks**
ADHD Adult	PTSD Adult	**Speech Tasks**	Caterpillar Passage	Word-color Stroop
	ADHD Adult	Open Response	CAPE-V	Productive Vocabulary
Speech Tasks			Open Response	Random Item Generation
Prolonged Vowel	**Speech Tasks**			Cinderella Story Retell
Maximum Phonation Time	Animal Fluency			
Rainbow Passage	Open Response			
Pitch Glides				
Loudness Task				
Free Speech				
Diadochokinesis				
Respiration				
Cookie Thief Description				
Grandfather Passage				

WORKFLOW: DEVELOPING A HYPOTHESIS-AGNOSTIC PROTOCOL



1. INPUT FROM CLINICAL EXPERTS
Collect comprehensive assessments, tasks, and measures across different disorders.



2. ORGANIZATION OF PART A
Create a shared Part A that can be compared across all diagnoses for consistency.



3. DISEASE-COHORT-SPECIFIC PART B
Develop Part B protocols for each disease cohort to provide more disease-specific diagnostic criteria.



RESULT: A hypothesis-agnostic protocol that balances consistency across participants with disease-specific depth for accurate diagnosis and analysis.

2 REVISIONS DURING OPTIMIZATION

PART A: All Participants	PART B: Mood Disorders	PART B: Respiratory Disorders	PART B: Voice Disorders	PART B: Neurological Disorders
About Me	**About Me**	**About Me**	**About Me**	**About Me**
Demographics	Mood-related Medical History	Respiratory Medical History	Voice Medical History	Neurological Medical History
Comprehensive Medical History	Mood-related Confounders	Respiratory Confounders	Voice Confounders	Neurological Confounders
Comprehensive Confounders				
Validated QOL Questionnaires	**Validated QOL Questionnaires**	**Validated QOL Questionnaires**	**Validated QOL Questionnaires**	**Other Tasks**
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Maximum Phonation Time	Animal Fluency			
Rainbow Passage	Open Response			
Harvard Sentences				
Pitch Glides				
Loudness Task				
Free Speech				
Diadochokinesis				
Respiration				
Cookie Thief Description x2				
Grandfather Passage				
Caterpillar Passage				
CAPE-Vr				
Frog Where Are You Story Recall				

● Added ● Eliminated ● Moved from Part B to Part A ● No Change

CHANGES MADE

ADDED (Newly Included)

- Harvard Sentences (Part A)
- Frog Where Are You Story Recall (Part A)

ELIMINATED (Removed)

- PTSD Adult (Mood Disorders)
- Animal Fluency (Mood Disorders)
- Voice Problem Severity (Voice Disorders)
- Winograd task (Neurological Disorders)
- Grandfather Passage (Part A)
- Rainbow Passage (Part A)
- Open Response (Respiratory Disorders)

MOVED (Relocated to Part A)

- ADHD Adult (Mood Disorders)
- Caterpillar Passage (Voice Disorders)
- CAPE-V (Voice Disorders)
- About Me (from all Part B sections)

CHANGE SUMMARY

+2	**Items Added** New assessments and tasks included
-10	**Items Removed** Duplicate or non-essential items removed
4	**Items Moved** Relocated to Part A for consistency
	GOAL Create a streamlined, consistent and comprehensive protocol across all parts.

Skills Check

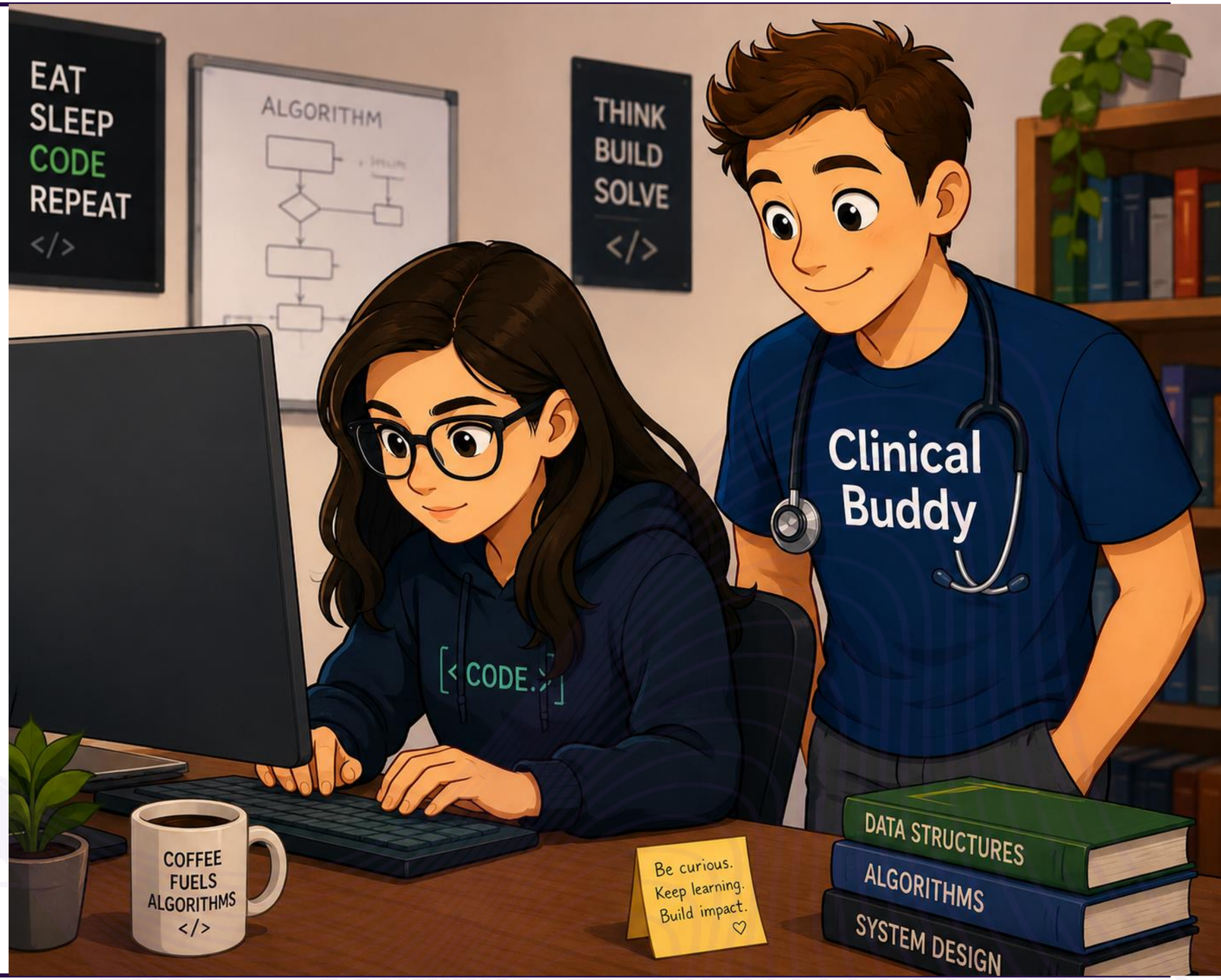
Phonetically balanced?

LCQ upper/lower airway?

Diadochokinesis?

Sustained phonation vs maximum phonation time?

PHQ-9 vs GAD-7 for depression?



Education

Join our Discord Channel: Hackathon 2026

https://discord.gg/NmkyQZhQ



B2AI VOICE

Canvas Learning Management System

Educational Content: As part of the registration, you will gain access to a library of online resources via a Canvas course, including educational videos and materials about the dataset, AI applications in voice data, and data governance.

https://wustl-catalog.instructure.com/courses/850

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Bridge2AI Resources

Visit this website for all of the B2AI resources in one place!

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Thank you to our Hackathon Sponsor!



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LET'S STAY CONNECTED!

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Team Formation



Post Assessment and Evaluation

Enter the Survey Access Code

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MY7RJ4YDA

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